

Project Summary

csci205_final_project

Project Details

Members

- Carlos Stolper
- Dipesh Bhattarai
- Jack Mclaud
- Lily Romano
- SooAh Lay

Project Retrospective

What was your initial goal?

Our initial goal was to create a user-friendly dashboard to help engineers and rocket enthusiasts track and understand rocket data. The project focused on building an interactive Java application that can import data files and visualize key rocketry metrics such as altitude, acceleration, velocity, trajectory, temperature, fuel levels, system power, and general sensor outputs. We aimed to make the dashboard easy to understand and visually appealing by using clear visuals like gauges, charts, maps, and status indicators. The goal was to help users quickly grasp important information and make sense of the rocket's performance through simple and intuitive design.

What did you achieve?

We successfully implemented all the features we planned for in this sprint. The dashboard displays real-time and imported data for key rocketry metrics, including accelerometer readings (accX, accY, accZ), gyroscope data (gyroX, gyroY, gyroZ), and orientation values such as yaw, roll, and pitch. It also shows environmental data like altitude, pressure, and temperature, as well as the rocket's position using longitude and latitude. A Map API is integrated to visualize the rocket's location, and a Weather API provides current weather conditions during the launch. We also added a video display feature to show live or recorded footage of the rocket launching. The dashboard is clean, easy to use, and visually intuitive, presenting data through charts, maps, and gauges. On the development side, we ensured the code is clean, well structured, and follows object-oriented programming principles, making it easy to maintain and expand in the future. Overall, the project meets all the technical and design goals we set at the beginning of the sprint.

What went well in the project?

We applied solid software engineering practices throughout the project, which helped us stay organized and efficient. By following object-oriented programming principles, we were able to write clean, modular, and maintainable code. We also adopted Scrum practices, clearly assigning roles, planning our sprints carefully, and holding regular meetings to track progress and adjust as needed. These methods helped us work

effectively as a team, stay focused on our goals, and meet our deadlines. Overall, our structured approach contributed to the success and quality of the final product.

What could be improved?

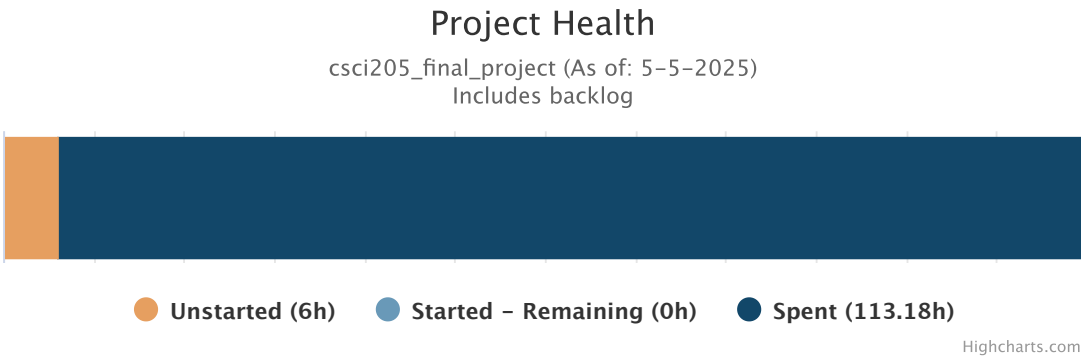
Communication There were still gaps in team communication, particularly regarding updates on individual progress and blockers. Improving transparency and consistent check-ins could help the team stay more aligned. Meeting Timing and Attendance Scheduling meetings that work for everyone is a challenge. Some meetings started late or had low attendance, which affected coordination. A more structured meeting schedule and better time management could help.

What would you change if you did the project again?

If we redid the project, we'd focus on making team communication better. Everyone would share quick updates daily or every few days what they're working on, any problems, and what's next. This keeps everyone on the same page without too many meetings. Sprint meetings would be a must, with clear plans and reminders so everyone shows up. If someone can't make it, they'd send an update first. We'd set clear rules for tools: email for big decisions, chats for fast questions, and a project tool for tasks, so there's no mix-up. After each sprint, we'd do a quick review to fix any communication issues. This would help us work together better and keep the project smooth.

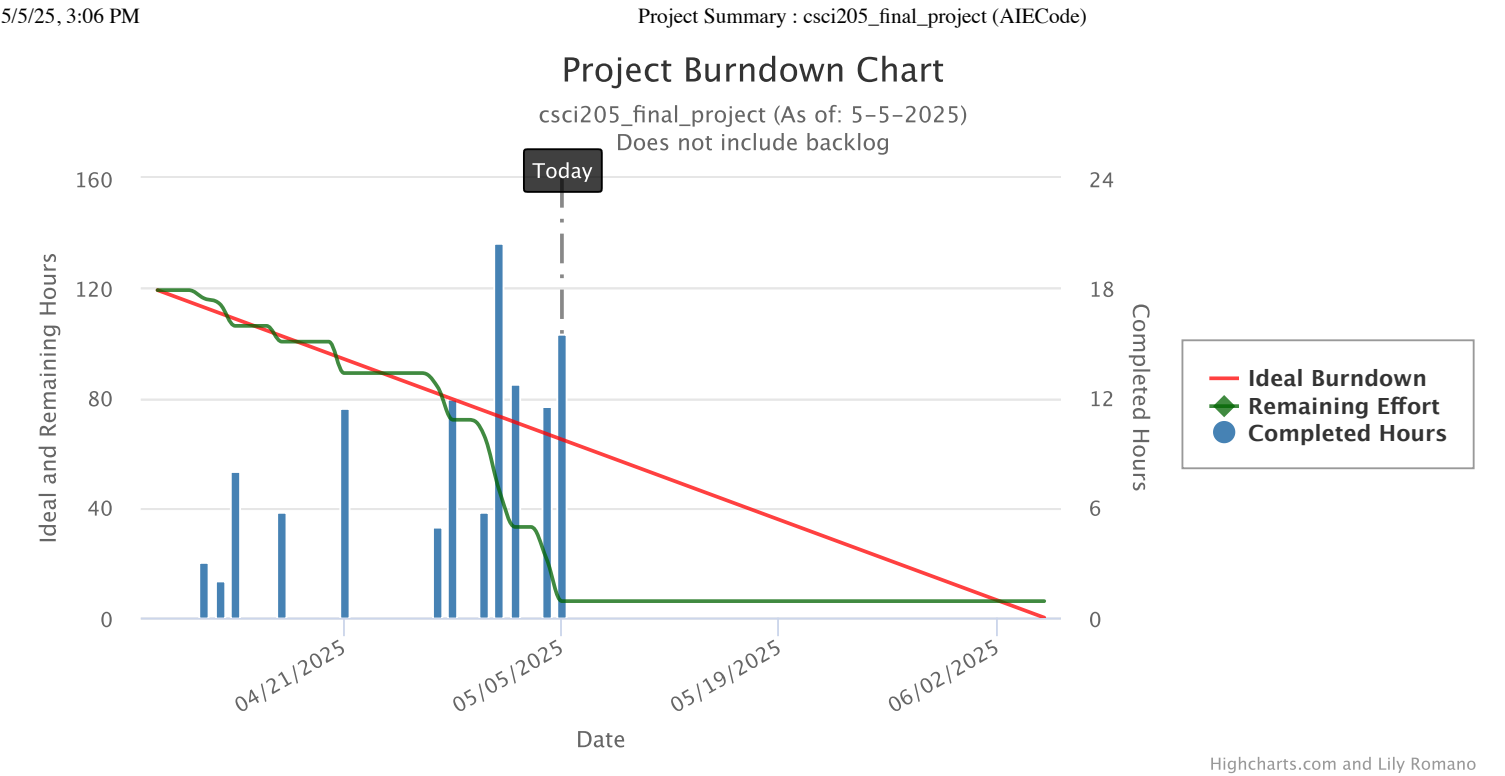
Charts

Health Bar



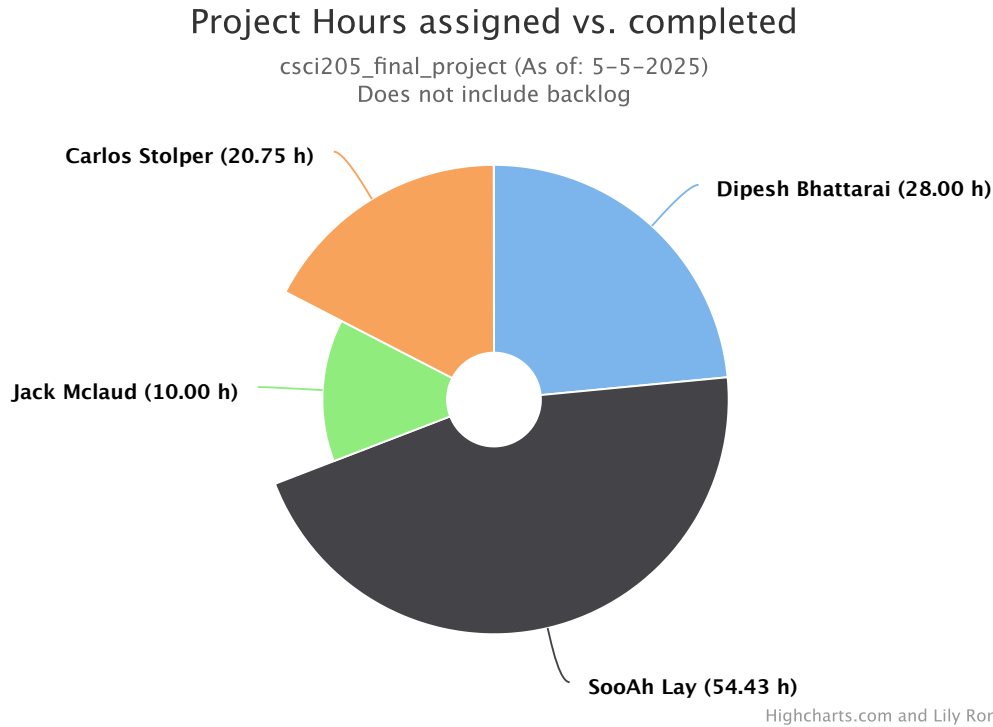
As of May 5, 2025, the csci205 final project has spent 113.8 hours, with 6 hours of work still unstarted and no hours marked as started but remaining, meaning we finished what we started and logged time consistently. We're very happy with the total work spent, as it reflects our steady effort.

Burndown Chart



The burndown chart for the final project shows that work was inconsistent at first, as we focused more on planning than implementation early on, with effort picking up significantly in the last two weeks. At the end we completed around 20+ hours of work, while the remaining effort dropped to nearly zero, reflecting a strong push to finish documents and implementation together.

Assignee Chart



The chart reflecting very well what each team member contribution and effort with different tasks assigned.

Name	User Stories	Bugs	Tech. Tasks	Design Tasks	Spikes	Doc.
Carlos Stolper	0	0	17.75	0	0	3
Dipesh Bhattarai	2	1	5	11	3	6
Jack Mclaud	0	0	10	0	0	0
SooAh Lay	8	2	29.93	6	3	5.5

Sprints

Sprint 1

Dates:

4-9-2025 to 4-14-2025

Review:**What went well in the sprint?**

One of the biggest successes of this sprint was that we clearly established each team member's role early on. This helped with communication and made it easier to assign tasks based on individual strengths. The division of responsibilities was balanced and efficient, allowing everyone to contribute meaningfully. All of the assigned tasks were completed on time, and we were able to manage our workload effectively without needing to rush at the last minute. Task assignments were well thought out, and we collaborated smoothly to ensure that each deliverable met our expectations. Overall, our team's organization, clear communication, and ability to stick to the timeline contributed to a successful and productive sprint.

What could be improved?

One thing could be improved is general communication. At times, there was a lack of clarity about certain updates or expectations, which led to a few small misunderstandings and delays. Improving our communication especially through more consistent check-ins or clearer documentation of decisions would help us stay more synchronized moving forward.

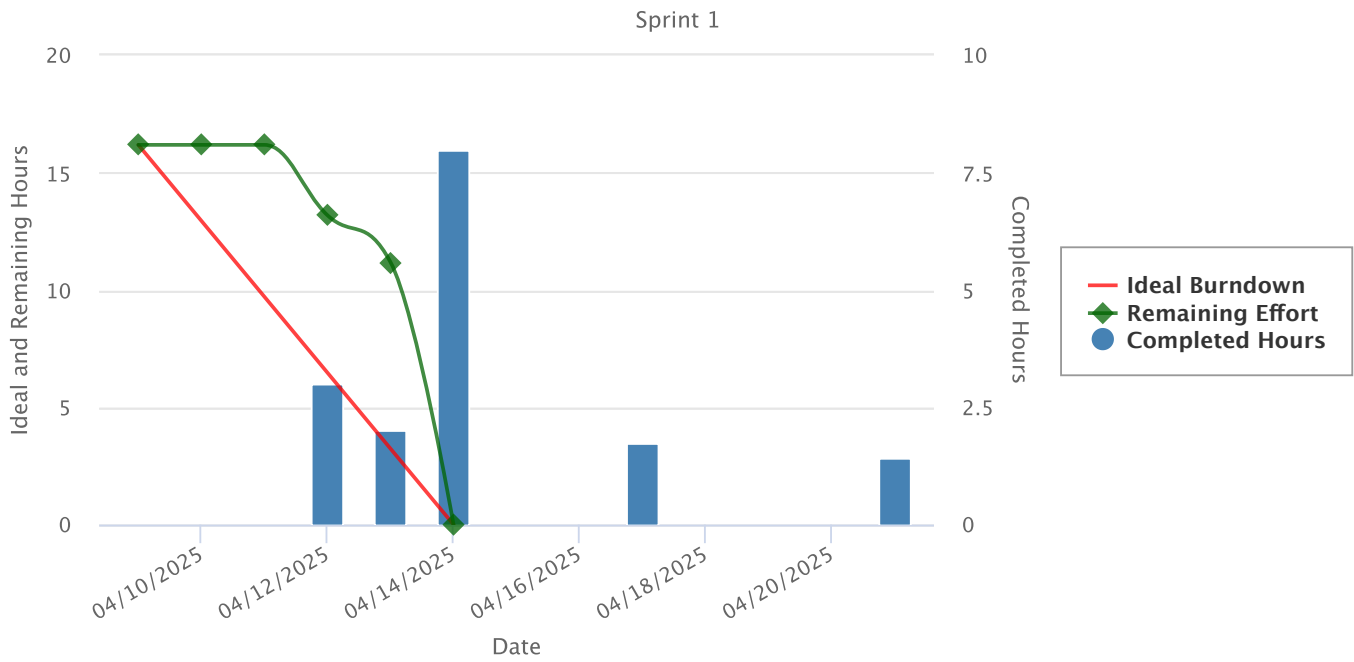
Are you on track? What is your plan if not?

Yes we are on track with everything.

What will you improve on in the next sprint?

Finding a proper/better meeting time for all of us that works, also having everyone in lecture. Unfortunately some of us can't make it to class lecture do to other class commitment.

Sprint Burndown Chart



Sprint 2

Dates:

4-14-2025 to 4-21-2025

Goal:

This sprint will be mostly technical implementation of each class unlike Sprint 1 which was more about documentation.

Review:

What went well in the sprint?

We did well with the technical parts of the project. Everyone finished their tasks on time. There was one task that took longer than expected, so we decided to move it to Sprint 3. Even so, we made good progress on it. We started getting the data and plotting it. Next, we'll focus on making the graph look better with proper visuals. Another success was finishing jam09. This was really helpful for our current sprint and will also make things easier in the next one.

What could be improved?

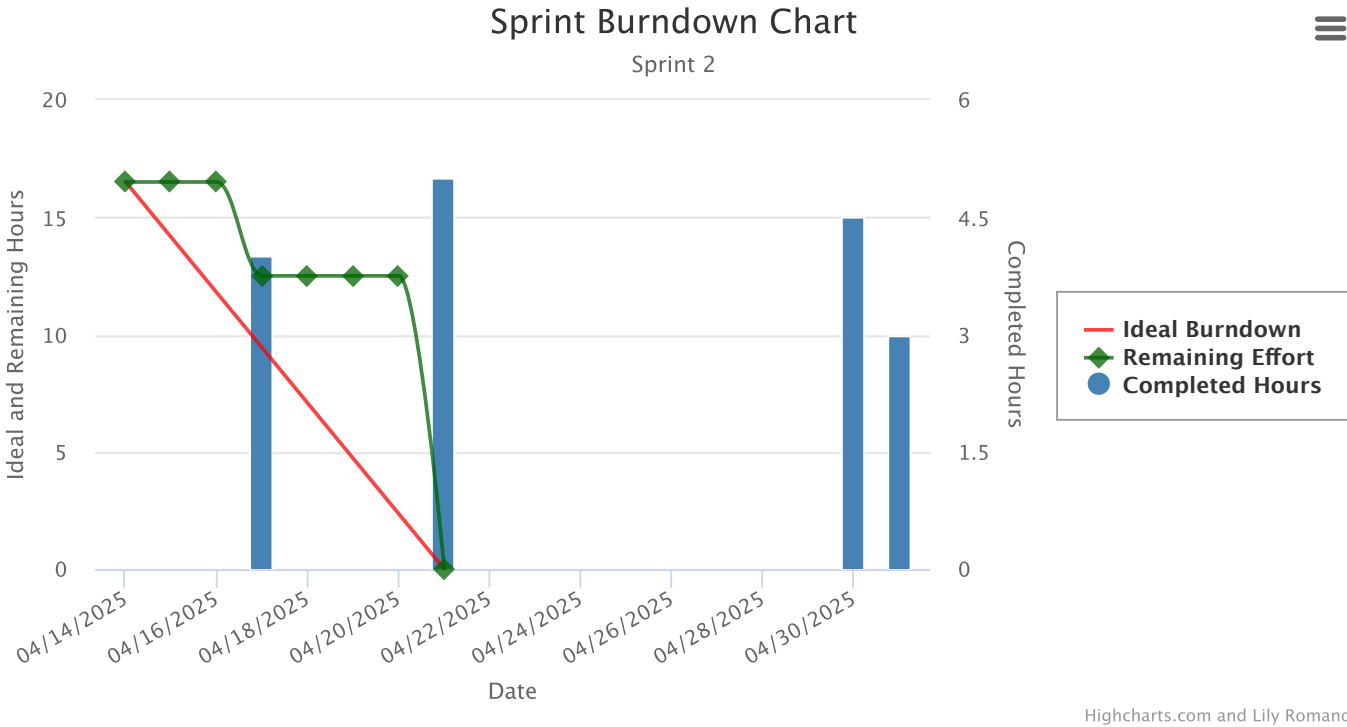
Meeting times are still a challenge for our team. It's hard to find a time that works for everyone, especially since some of us have classes at the same time. Because of this, we weren't always able to meet together, and it made communication harder. Also, texting isn't always the best way to talk about tasks. Sometimes it caused confusion or delays.

Are you on track? What is your plan if not?

We're about 80% on track. Most of the tasks were completed, but one task had to be pushed to the next sprint because it took more time than expected. So we plan to improve how we estimate the time needed for each task. This will help us stay on schedule and manage our workload better in the next sprint.

What will you improve on in the next sprint?

We plan to improve our teamwork by setting aside one or two hours on the weekend to meet. This will help make up for the time we couldn't meet during class.



Sprint 3

Dates:

4-21-2025 to 4-28-2025

Goal:

This sprint will now focus on UI and data visualization using SceneBuilder to help

Review:

What went well in the sprint?

The Weather API was successfully implemented and integrated into the project. All necessary files were written, properly structured, and are functioning as intended. Bugs and issues during development were quickly identified and resolved. Code quality and documentation were kept at a high standard, making it easier for future updates.

What could be improved?

Scheduling meetings that work for everyone remains a challenge. Some meetings started late or had low attendance, which affected coordination.

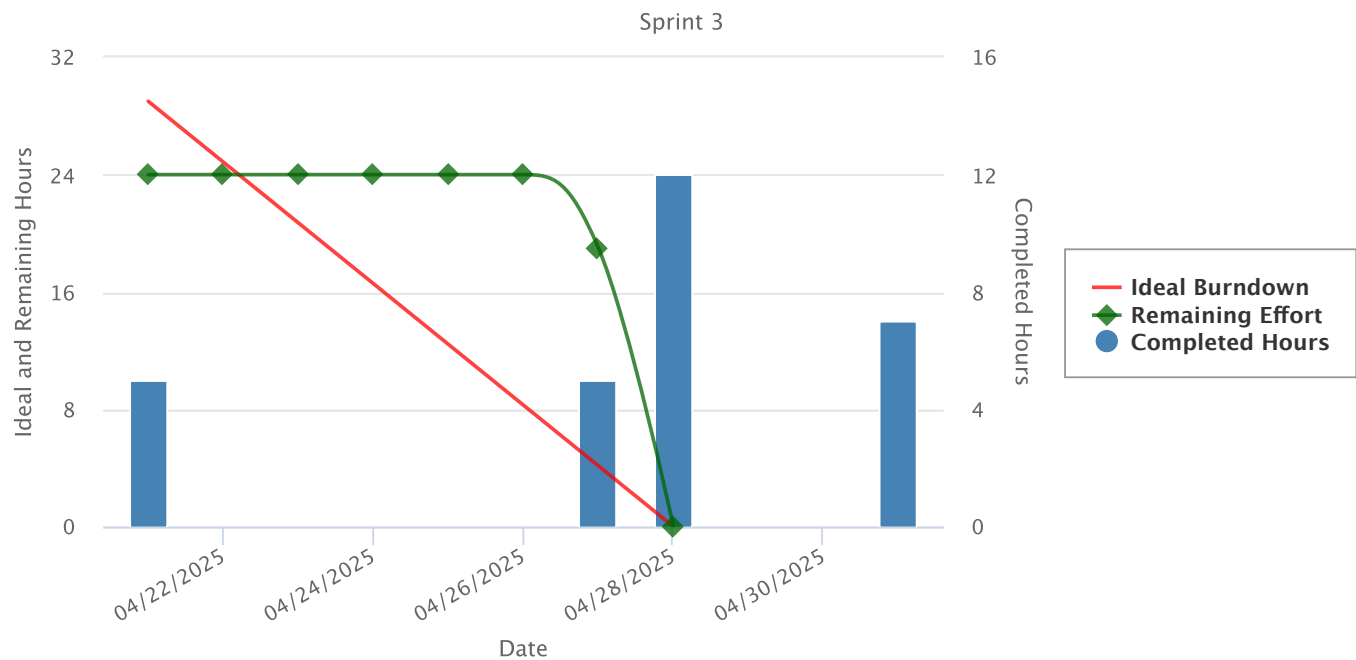
Are you on track? What is your plan if not?

In terms of the sprint tasks, we are on track. All planned deliverables are progressing as scheduled

What will you improve on in the next sprint?

Make sure everyone puts in equal effort. Split the work more fairly. Check in more often to stay on track.

Sprint Burndown Chart



Sprint 4

Dates:
4-28-2025 to 6-5-2025

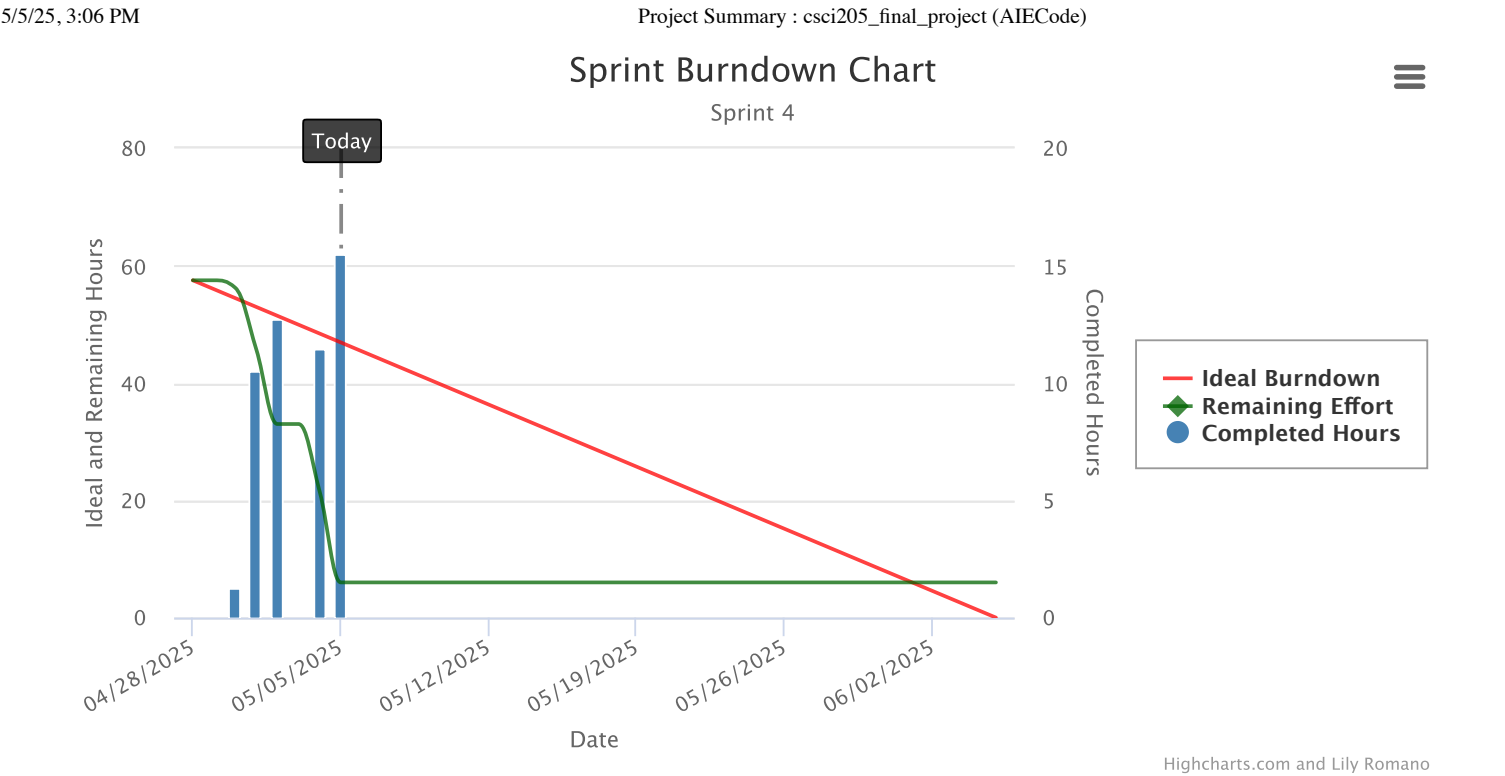
Goal:
Wrap up the project with good standard code quality and documentation

Review:
What went well in the sprint?

All sprint tasks were successfully implemented, and the team stayed on track with the planned timeline. One of the key highlights was the development of the dashboard, which turned out very well. It not only met the initial design and functionality requirements but also incorporated valuable user feedback along the way, resulting in a clean, intuitive interface with efficient data visualization. The integration between components was smooth, and performance showed significant improvements.

What could be improved?
Even though we finished all the tasks, teamwork could have been better. There were times when team members didn't communicate clearly or weren't on the same page. This caused some confusion and extra work. Some people worked alone without checking in with others, which made it harder to share ideas or give feedback. To improve, we can try having more regular check-ins, make sure everyone knows their tasks, and use tools to keep track of what's going on.

If you were to continue the project, what would you improve on in the next sprint?
This is the final sprint.



Personas



Storm Hansen

Quote

'Looking at all of these interactive and dynamic charts, I feel a certain way I cannot describe... mmph~~'

Narrative

Storm is an experienced data analyst that specializes in post-flight analysis: aggregating dozens of launches based on rocket type, running statistical and inferential tests, and creating performance reports.



Konstantin Lehen

Quote

"When I am out on the rocket pad, I need to see any discrepancies or anomalies within mS -- there cannot be a lapse in judgement!"

Narrative

Kon (short for Konstantin) works for a company that frequently contracts with the government; he oversees each launch's safety (and or risk) and performance. He must correlate sensor readings and parameters, e.g. alt., accel., pressure, etc, with live-video and countdown information to make a decision whether it is good or not to go.



Sara Harju

Quote

"Watching a launch while I track every data point in real time--that's how I learn what really happens under the hood."

Narrative

Sara is on the design team for propulsion systems and needs to correlate (and compare) live-streamed sensor information (thrust, pressure, temp) with launches she is simulating. She plans to build her own sub-orbital vehicle someday; she utilizes this dashboard as a learning tool and benchmark



Ramesh Gupta

Quote

"I need to evaluate and compare the chamber's pressure, temperature and fuel flow-rates, to ensure that everything is a go"

Narrative

Ramesh is a graduate student at a large state-run research university where he runs the engine test-stand group. He performs tasks like static-fire tests and different injector settings and propellant mixes. His main focus is on the trends across different burn mixtures.



Jenny Ramos

Quote

"I love diving into Physics -- especially Newtonian notion -- and the material side of rocketry. I love figuring out how each weld, component, and composite layer behaves under the stress while in air!"

Narrative

Jenny is a graduated mechanical engineer, with an interest in aerospace engineer as a hobby. She likes to design and print out rocket figurines for herself and her friends. She also builds and launches small-scale



Sofus Løvdal

Quote

"Working behind the scenes gives me a sense of thrill, but also despair! I rely on a unified view--video, telemetry, countdowns, and communications--so that everything stays in sync"

Narrative

Sofus staffs the mission control desk during launches, where he oversees different metrics to confirm that everything is a go. He needs a clean and synchronized

rockets in her free time, helping other enthusiasts as well. She browses online forums and participates in a thriving community.

dashboard so that he, the engineering, safety, inspection, and launch team are in sync and in sync.



Sebastian May

Quote

"I want a rock-solid communication platform between the front & backend services, and any other crucial microservice there is. I do not want any surprises and want everything there!"

Narrative

Seb (Sebastian) builds and maintains software and data pipelines that take raw telemetry and other data for the dashboard and develops real-time messaging between different end systems. He needs and loves well-documented code and clean error handlings, so everything works as needed. He is very detailed oriented and checks every commits, pushes, and merges.



Susan Caldwell

Quote

"I've always loved rockets, but never got a chance to build, test, and deploy one until later in life! I work on my own Rockets and with the local high school to teach kids the procedures and process of basic rocketry!"

Narrative

Susan is a hobbyist who builds and flies small rockets on their large farmland on weekends. She records his rocket data onto a microSD card and has it in a .CSV format. and wants a way to extract, visualize, and compose the data into a readable view. She also wants the local students to learn about data processing and visualization for a clean launch.

Table of Work

Showing 1 to 48 of 48 entries

Search:

Title	Type	Est.	Spent
Opened (2)		6 h	0
Sprint 4 (2)		6 h	0
model package JUnit Test	Technical Task	3 h	0
Re-organize/design UML	Documentation	3 h	0

Title	Type	Est.	Spent
Closed (38)		113 h, 11 m	113 h, 11 m
Sprint 1 (6)		16 h, 11 m	16 h, 11 m
CRC Cards	Documentation	2 h	2 h
Create Classes	Technical Task	4 h, 21 m	4 h, 21 m
Creating Test Cases	Technical Task	1 h, 50 m	1 h, 50 m
Mock-up User Interface	Design Need	2 h	2 h
README.md file	Documentation	1 h	1 h
UML diagrams	Technical Task	5 h	5 h
Sprint 2 (5)		16 h, 30 m	16 h, 30 m
CountDown Class	Technical Task	3 h	3 h
Sensor Class	Technical Task	3 h	3 h
Start / Get through Jam09	Spike	6 h	6 h
System Operation Class	Technical Task	1 h, 30 m	1 h, 30 m
Timeline class	Technical Task	3 h	3 h
Sprint 3 (9)		29 h	29 h
Block design based off of Initial GUI design	Design Need	4 h	4 h
Button activation for graphs and graphic	User Story	2 h	2 h
Dynamic graph for Gyro data	Technical Task	4 h	4 h
LiveFeed	Technical Task	4 h	4 h
Location Class	Technical Task	3 h	3 h
Pressure and Temperature Visual Graph	Technical Task	4 h	4 h
Removing plotted dots on graphs	Technical Task	1 h	1 h
System Operation Class	Technical Task	2 h	2 h
Weather API	Technical Task	5 h	5 h
Sprint 4 (18)		51 h, 30 m	51 h, 30 m
Controller and View JUnit Tests	Technical Task	3 h	3 h
Creating and adding more User Personas	Design Need	2 h	2 h
Design Manual	Documentation	4 h	4 h
Dynamic graph for all three graphs	Technical Task	4 h	4 h

Title	Type	Est.	Spent
Google Map API	Technical Task	3 h	3 h
Implement status of the rocket state	User Story	3 h	3 h
Implement the GPS location of the Rocket on dashboard	User Story	5 h	5 h
Implement Velocity graph on UI	Technical Task	4 h	4 h
Impletement Pressure and Temperature visualization	Technical Task	4 h	4 h
Major UI Revamp	Design Need	6 h	6 h
Reconfigure tasks to appropriate type	Documentation	2 h	2 h
Scrum Report	Documentation	1 h	1 h
Speed of the Dynamic Graphs (too fast)	Bug	3 h	3 h
Update and robust-ify existing User Personas	Design Need	2 h	2 h
Update ReadMe.md	Documentation	1 h	1 h
Update Sara Harju User Persona	Design Need	1 h	1 h
Use Case Diagram	Documentation	30 m	30 m
User Mannual	Documentation	3 h	3 h
Trashed (8)		0	0
Sprint 2 (1)		0	0
Block design based off of Initial GUI design	Design Need	0	0
Sprint 3 (1)		0	0
Dynamic graph for Gyro data	Technical Task	0	0
Sprint 4 (2)		0	0
Implement status of the rocket state	Technical Task	0	0
Implement the GPS location of the Rocket on dashboard	Technical Task	0	0
Backlog (4)		0	0
AeroSpace Engineer Story(Sara Harju)	User Story	0	0
Creating and adding more User Personas	Documentation	0	0
UML diagrams	Technical Task	0	0
UML diagrams	Technical Task	0	0

Daily Scrum

Daily Scrum Notes

04/14/2025

1. Jack

- Accomplished: UML diagram.
- Working: implementation Countdown class
- Challenges: Draw.io is not user friendly.

2. Carlos

- Accomplished: project structure, create file for all the classes.
- Working: implementation on Timeline class.
- Challenges: Setting up the project structure, package or module?

3. SooAh

- Accomplished: Dashbord Design, CRC cards, README.md
- Working: implementation Sensor class
- Challenges: reading data from csv.

4. Dispesh

- Accomplished: Initialize gradle, user story, skeleton template
- Working: fix the set up structure and gradle.
- Challenges: setting up gradle.

04/17/2025

1. Jack

- Accomplished: Countdown class 50%
- Working: Countdown class
- Challenges: Busy with other classes

2. Carlos

- Accomplished:
- Working: implementation on Timeline class.
- Challenges: git command

3. SooAh

- Accomplished: Sensor class
- Working: SceneBuilder
- Challenges: reading data from csv.

4. Dispesh

- Accomplished: Initialize gradle, user story, skeleton template
- Working: fix the set up structure and gradle.
- Challenges: setting up gradle.

04/24/2025

1. Jack

- Accomplished: Learned API
- Working: Trying implement API
- Challenges: Figuring out how to do it.

2. Carlos

- Accomplished: Finished GPS class
- Working: implementation on csv
- Challenges: how to implement csv, liveFeed

3. SooAh

- Accomplished: Get the data to display on SceneBuilder
- Working: Fixing the disproportionate on SceneBuilder
- Challenges: fxml and scenebuilder

4. Dispesh

- Accomplished: animation
- Working: getting API
- Challenges: finding the API